TASK 2 · RESEARCH REPORT

LLM Prompt Engineering for RASE Analysis

§47 Aufenthaltsräume · Musterbauordnung (MBO)

Mohammed · RASE Research Project · Hochschule Merseburg · 2026

1. Introduction

The objective of this task was to investigate whether a Large Language Model (LLM) can reproduce a human-generated RASE annotation of a legal regulation. In Task 1, a manual RASE mark-up of §47 Aufenthaltsräume from the Musterbauordnung (MBO) was created using the ACCORD Rule Formalization Tool. The annotation distinguishes between coloured BOXES (Sections, which group text) and coloured TAGS (Spans, which mark the individual testable metrics inside a box).

Main Research Question: Can an LLM reproduce a human-level RASE annotation — including the correct Box/Tag structure — through iterative prompt engineering?

2. Manual RASE Analysis (Task 1)

RASE uses four colours. Every colour can appear as a BOX (a Section that wraps a group of words) and as a TAG (a Span that marks the precise metric inside a Section):

Colour	Letter	Box (Section)	Tag (Span / Metric)
Blue	R – Requirement	Wraps a sentence that imposes an obligation	Marks the measurable value, e.g. `>= 2.40 m`
Green	A – Application	Wraps the subject of the rule	Marks the object, e.g. `:type == :Aufenthaltsraeume`
Purple	S – Selection	Wraps a conditional branch	Marks the condition word (wenn / oder)
Orange	E – Exception	Wraps an exempting clause	Marks the exempting condition / value

Each TAG is additionally one of three types:

Tag type	Meaning	data-raseproperty format	Extra fields
Object	A thing or subject (no number)	`:type == :Name`	—
Property	A measurable value (a number)	`:name OPERATOR value :unit`	Operator (>=, <=, >, <, ==), Value (number), Unit
Reference	A reference to another clause	cross-reference to a clause	—

In §47 Abs. 1 all the green Application tags are OBJECT tags, and all the blue Requirement tags that carry a number are PROPERTY tags. For every Property tag we entered an operator, a value and a unit in the ACCORD tool, e.g. *"mindestens 2,40 m"* became operator >=, value `2.40`, unit `metres`.

The final structure of §47 Abs. 1 (after correction by the supervisor, Nick Nisbet) is a single overall Requirement Section that contains one Requirement Section followed by three Exception Sections:

2.1 — Box and Tag Structure (exactly as marked up)

# Box (Section)	Tags inside (type)	data-raseproperty (operator / value / unit)
Overall R-BOX (wraps the whole §47 Abs. 1)	— (container only)	—

Satz 1 R-BOX
(Requirement Section)
	A-TAG "Aufenthaltsräume" (Object)
	R-TAG "mindestens 2,40 m" (Property)
		`:type == :Aufenthaltsraeume`
		`:lichteRaumhoehe >= 2.40 :metres`
		`(>= / 2.40 / metres)`

# Box (Section)	Tags inside (type)	data-raseproperty (operator / value / unit)

Satz 2 E-BOX
(Exception Section)
	A-TAG "Aufenthaltsräume" (Object)
	A-TAG "im Dachraum" (Object)
	R-TAG "mindestens 2,20 m" (Property)
	R-TAG "...der Hälfte ihrer Netto-Raumfläche" (Property)
		`:type == :Aufenthaltsraeume`
		`:type == :imDachraum`
		`:lichteRaumhoeheDachraum >= 2.20 :metres`
		`(>= / 2.20 / metres)`
		`:NettoRaumflaeche >= 0.5 :metres`
		`(>= / 0.5 / metres)`

Raumteile ... 1,50 m NO BOX — NOT marked up	(none)	Measurement methodology, not a requirement (see 2.3)

Satz 3 E-BOX
(Exception Section)
	R-TAG "Wohngebäuden" (Object)
	R-TAG "Gebäudeklassen 1 und 2" (Property)
		`:type == :Wohngebaeude`
		`:Gebaeudeklasse <= 2`
		`(<= / 2 / none)`

2.2 — Why is Satz 2 (attic rooms) an EXCEPTION and not a Requirement?

This was the key correction from the supervisor. At first the attic-room sentence was marked up as a second Requirement Section (R-BOX). It must instead be an Exception Section (E-BOX). The reasoning:

Example: Imagine an attic room with a ceiling height of 2,30 m. Tested against Satz 1 (which demands >= 2,40 m) the room would FAIL. But the legislator does not want to forbid such an attic room. Satz 2 gives it a SECOND, easier path: a height of >= 2,20 m over at least half of the floor area.

Because Satz 2 lets a room pass that would otherwise fail the general rule, it functions as an EXCEPTION to Satz 1 (*lex specialis* — the special rule overrides the general rule). Therefore it is an E-BOX, not an R-BOX.

2.3 — Why is the 1,50 m sentence NOT marked up at all?

“Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht.”

At first this looked like an Exception that excludes low rooms (e.g. cupboards). After careful reading with the supervisor it became clear that the sentence does NOT impose or exempt any obligation. It only DEFINES how the net floor area is measured: when checking whether >= 2,20 m is reached over “at least half” of the area, any floor portion lower than 1,50 m is simply removed from the area calculation. The sentence does not prohibit such areas — it only changes the arithmetic. It is therefore MEASUREMENT METHODOLOGY and receives no RASE mark-up. A keyword-based reader would wrongly tag it as an Exception; correct semantic reading leaves it unmarked.

2.4 — The clause title is an Application

The supervisor also pointed out that the title word “Aufenthaltsräume” in the heading of §47 is itself an Application: it defines the subject that the whole clause applies to. The word “Aufenthaltsräume im” (Dachraum) in sub-clause 2 must therefore be understood as a KIND of Aufenthaltsraum — which is exactly why sub-clause 2 forms a further Exception Section rather than an independent rule.

3. HTML Mark-up Specification

The supervisor provided the exact HTML rules that the mark-up (and the LLM output) must follow. These rules were built into the final prompt (V3):

Rule 1. Do not move, remove or change the given text — only wrap parts of it in tags.

Rule 2 — single metric (TAG). Where there is a single word or phrase, add a `<span>` with `data-rasetype` set to `'Requirement'`, `'Application'`, `'Selection'` or `'Exception'`; a second attribute `id` that is unique and starts with `'m'`; and a third attribute `data-raseproperty` giving the meaning.

Rule 3 — group (BOX). Where several mark-ups are grouped, add a `<div>` with `data-rasetype` set to `'RequirementSection'`, `'ApplicationSection'`, `'SelectionSection'` or `'ExceptionSection'`; and a second attribute `id` that is unique and starts with `'S'`.

3.1 — Example of the exact tag syntax used

TAG (span):

```
<span data-rasetype="Requirement" id="m2" data-raseproperty=":lichteRaumhoehe >= 2.40 :metres">mindestens 2,40 m</span>
```

BOX (div):

```
<div data-rasetype="RequirementSection" id="S1"> ... </div>
```

4. Methodology

Three prompt versions were developed and tested against the manual Box/Tag annotation. Each prompt used the same regulation text and became progressively more sophisticated.

Version	Purpose
V1	Basic RASE extraction — keyword-driven identification
V2	Legal interpretation — semantic reasoning + Box/Tag idea
V3	Human expert — full HTML spec, tag types, lex specialis

Outputs were evaluated on six criteria: correct RASE classification, correct Box vs. Tag structure, correct tag type (Object / Property / Reference) with operator-value-unit, legal interpretation quality, recognition of the attic-room exception (*lex specialis*), and correct handling of the 1,50 m measurement sentence.

5. Prompt V1 — Basic RASE Extraction

V1 simply asked the LLM to identify Requirements, Applications, Selections and Exceptions, with no legal-interpretation guidance and no instruction about Box/Tag structure.

Results — what V1 got right

✓ Identified Requirement R1 (>= 2.40 m)
✓ Identified Requirement R2 (attic, >= 2.20 m)
✓ Identified Application A1 (Aufenthaltsräume)
✓ Identified Application A2 (im Dachraum)
✓ Identified Exception E1 (Gebäudeklassen 1 und 2)

Issues Found

Problem 1 — 1,50 m sentence wrongly classified. V1 tagged *“Raumteile ... bis zu 1,50 m”* as a Selection. The human mark-up leaves it unmarked because it is only measurement methodology (see 2.3).

Problem 2 — no *lex specialis* recognition. V1 treated Satz 1 and Satz 2 as two independent Requirement Sections (two R-BOXes). It did not recognise that Satz 2 is an Exception Section overriding Satz 1 (see 2.2).

Problem 3 — no Box/Tag distinction. V1 produced a flat list of elements and never expressed which items are Sections (Boxes) and which are the metric Tags inside them.

Evaluation: Strong keyword extraction, weak legal reasoning. The model matched words, not meaning.

6. Prompt V2 — Legal Interpretation

V2 added legal-reasoning instructions: think like a legal expert, explain the meaning of each sentence, distinguish requirements from measurement rules, and explain when a special case overrides a general rule.

Improvements

The output improved clearly. V2 recognised that the attic room is a SPECIAL case and described the relationship between Satz 1 and Satz 2, introducing the concept of *lex specialis*. It began to treat Satz 2 as an exception-like special rule rather than a fully independent requirement.

Remaining Problems

The 1,50 m sentence was still forced into a category (Exception). The model still tried to give every sentence a RASE element instead of leaving the measurement sentence unmarked.

Evaluation: Clear gain in semantic reasoning, but the boundary between a true legal rule and supporting measurement logic was still unresolved.

7. Prompt V3 — Human Expert Interpretation

V3 added explicit expert-review principles: legal-hierarchy analysis, *lex specialis* reasoning, and the key instruction that NOT every sentence must become a RASE element. It also asked the model to separate measurement methodology from real obligations.

Results — V3 matched the manual mark-up

✓ Satz 1 correctly identified as the general Requirement
✓ Satz 2 correctly identified as the attic-room special rule (Exception / *lex specialis*)
✓ Satz 3 correctly identified as the GK 1 & 2 Exception
✓ Relationship between Satz 1 and Satz 2 correctly recognised
✓ The 1,50 m sentence correctly identified as measurement methodology — left unmarked
✓ Legal hierarchy (general rule → special rule → exception) correctly mapped

Evaluation: V3 achieved near-human performance and reproduced the same logical structure (one Requirement followed by three Exceptions) as the manual annotation.

8. Comparison of Results

Feature	Human	V1	V2	V3
Requirement R1 (>= 2.40 m)	✓	✓	✓	✓
Requirement R2 (attic >= 2.20 m)	✓	✓	✓	✓
Application A1 (Aufenthaltsräume)	✓	✓	✓	✓
Application A2 (im Dachraum)	✓	✓	✓	✓
GK1/GK2 Exception	✓	✓	✓	✓
Lex specialis (Satz 2 = Exception)	✓	✗	✓	✓
1.50 m = Measurement (unmarked)	✓	✗	✗	✓
Box / Tag structure	✓	✗	partly	✓
Human-Level Reasoning	High	Low	Medium	High

9. Compliance Test (Cases A, B, C)

To check whether the LLM applies the mark-up correctly, the model was asked to decide compliance for three concrete rooms. This tests whether the rules (R1, R2, the attic exception, and the GK 1/2 exception) are applied as a human would apply them.

Case	Facts	LLM decision	Agree?
A	Normal habitable room, height 2.50 m, building class 4	**PASS** — meets R1 (>= 2.40 m)	Yes
B	Attic room, height 2.30 m over 60% of the floor area, building class 4	**PASS** — fails R1 but claims the attic Exception (>= 2.20 m over >= half)	Yes
C	Habitable room, height 2.10 m, residential building class 2	**PASS** — R1 and R2 do not apply (GK 1/2 Exception)	Yes

In all three cases the LLM (using the V3 prompt) reached the same decision a human expert would. Case B is the important one: it shows the model correctly treats the attic provision as an EXCEPTION — the room fails the general 2.40 m rule but still passes by claiming the attic exception. Case C confirms the model applies the building-class exemption correctly.

10. Key Findings

Legal Interpretation is Harder than Extraction

The LLM could identify Requirements and Exceptions from the start. Understanding the legal RELATIONSHIP between rules — and the Box/Tag hierarchy — required several prompt refinements.

Measurement Methodology Causes Confusion

The hardest sentence was *“Raumteile ... bis zu 1,50 m bleiben außer Betracht.”* The model repeatedly tried to classify it as a RASE element; only V3 correctly left it unmarked as measurement methodology.

Lex Specialis Requires Explicit Guidance

Without explicit instructions the model could not see that the attic-room provision (Satz 2) is an Exception that replaces the general rule (Satz 1). Explicit legal-reasoning guidance was necessary.

Prompt Engineering Has a Significant Impact

Each iteration improved the output. The path from V1 to V3 shows that prompt design is a major factor in reaching human-level results.

11. Conclusion

The experiment demonstrated that Large Language Models can perform RASE analysis with a high degree of accuracy when guided by a well-engineered prompt.

Prompt	Key Outcome
V1	Extracted the basic structure but lacked legal understanding and Box/Tag distinction.
V2	Improved semantic reasoning and identified the special relationship between the rules.
V3	Reached near-human performance: correct hierarchy, correct exception handling, and the 1,50 m sentence correctly left unmarked.

The results suggest that LLMs can effectively support regulatory rule formalisation. However, expert review remains necessary to validate complex legal interpretations and resolve ambiguities such as the attic-room exception and the measurement sentence.

12. Next Step — Task 3

Based on these results, the next phase is the development of an LLM-supported RASE extraction application. The final V3 prompt will be integrated into the application and tested against additional regulations and examples created by other groups.

Goal: Evaluate whether the model can consistently reproduce human-generated RASE annotations — with correct Box/Tag structure — across multiple clauses, and whether prompt engineering can serve as a reliable foundation for automated rule formalisation.